FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – III

JEE (Main)-2019

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. A

Sol. $a_t = 0.4\ m/s^2$

$a_c = \frac{F}{m}$

$a_c = \frac{mv^2}{mr}$

$a_c = \frac{v^2}{r} \Rightarrow \frac{6^2}{120} \Rightarrow 0.3\ m/s^2$

$a_{net} = \sqrt{a_c^2 + a_t^2} = \sqrt{(0.3)^2 + (0.4)^2} = 0.5$

2. C

Sol. $4y + kx = 3$

$4y = 3 - kx$

$y = -\frac{k}{4}x + \frac{3}{4}$

slope $m_1 = \frac{k}{4}$

$F \Rightarrow 3\,\hat{i} + 4\,\hat{j}$

slope $m_2 \Rightarrow \frac{4}{3}$

for perpendicular $m_1 \times m_2 = -1$

$-\frac{k}{4} \times \frac{4}{3} = -1$

$k = 3$

3. D

Sol. By parallel axis theorem

$I = M\left(\frac{R^2}{4}+\frac{\ell^2}{3}\right)$

$= M\left[\frac{R^2}{4}+\frac{(6R)^2}{3}\right]$

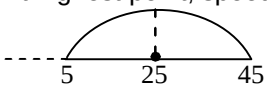
$I = \frac{49}{4}MR^2$

4. D

Sol. 1134.04g

5. D

Sol. At highest point, speed is not zero and acceleration is g downward.



6. B

Sol. T-Mg = ma

7. B

Sol. $f_x = \frac{dU}{dx} = \frac{d}{dx}(6x + 8y) = 6$

$f_y = \frac{dU}{dy} = 8$

$a_x = \frac{6}{2} = 3,\ a_y = \frac{8}{2} = 4$

after time (t) $x = x_0 - \frac{1}{2}a_x t^2 = 6 - \frac{1}{2}\times 3 \times t^2$

$y = y_0 - \frac{1}{2}a_y t^2 = 4 - 2t^2$

body crosses y axis at x = 0

$6 - \frac{3}{2}t^2 = 0,\ t = 2$ sec

8. A

Sol. $mv\cos\theta = \frac{m}{2}(-v\cos\theta) + \frac{m}{2}v'$

$v' = 3v\cos\theta$

9. C

Sol. At highest point

$v = u\cos 60^\circ = \frac{u}{2}$

Hence at maximum height perpendicular distance from line of projection

$= \frac{u^2\sin^2\theta}{2g} = \frac{3u^2}{8g}$

$L = mvr = m(u\cos 60^\circ)\times\frac{3u^2}{8g} = \frac{3mu^3}{16g}$

10. A

Sol. $mg - T = ma$

$\tau = mg.\frac{\ell}{2}$

$\alpha = \frac{\tau}{I} = \frac{3g}{2\ell}$

$a = \frac{L\alpha}{2}$

$mg - T = \frac{3mg}{4}$

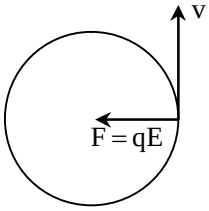
$T = \frac{mg}{4}$

11. C

Sol. Draw free body diagram & apply equation of motion.

12. A

Sol. Here electric force on the particle provide required centripetal force.



$F = qE = \frac{mv^2}{r}$

$\frac{q\lambda}{2\pi\varepsilon_0 r} = \frac{mv^2}{r}$

$v = \sqrt{\frac{\lambda q}{2\pi\varepsilon_0 m}}$

$T = \frac{2\pi r}{v}$

$T = 2\pi r\sqrt{\frac{2\pi\varepsilon_0 m}{\lambda q}}$

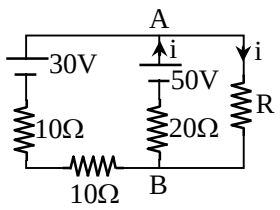
13. C

Sol. As there is no current through 30 V cell

Hence, $V_{AB} = 30\ V = iR$

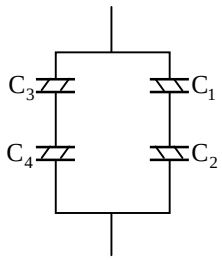
and $i = \frac{50-30}{20} = 1A$

Hence, $R = 30\ \Omega$



14. A

Sol. $C = \frac{\in_0 W}{d}$



$C_1 = \frac{6\in_0 W}{d}$; $C_2 = \frac{4\in_0 W}{d}$

$C_3 = \frac{6\in_0 W}{d}$; $C_4 = \frac{2\in_0 W}{d}$

$C_{eq} = \frac{C_1C_2}{C_1+C_2} + \frac{C_3C_4}{C_3+C_4}$

15. D

Sol. When only capacitor is removed, the circuit will be RL circuit and tan $\phi_1 = \frac{X_L}{R} = \sqrt{3}$

or $X_L = 100\sqrt{3}\ \Omega$

When only inductor is removed then

tan $\phi_2 = \frac{X_C}{R} = \sqrt{3}$

or $X_C = 100\sqrt{3}$

In LRC circuit

$Z = \sqrt{R^2 + (X_L - X_C)^2}$ = 100 Ω

$i = \frac{V}{Z}$ = 2A

$P = i^2R$ = 400W

16. A

Sol. Let these are N turns

hence $(2\pi r)N = \ell$

$r = \frac{\ell}{2\pi N}$

τ = iAB sin θ

$\tau = \frac{i\ell^2 B}{4\pi N^2}\sin\theta$

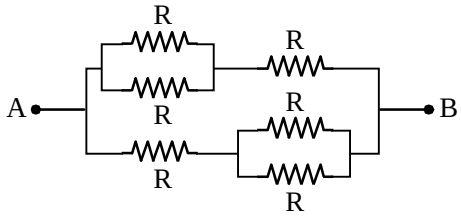
for τ_{max}, θ = 90° and N = 1

i.e., $\tau_{max} = \frac{i\ell^2 B}{4\pi}$

17. A

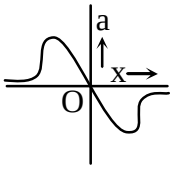
Sol. The given circuit can be simplified as

$R_{eq} = \frac{3}{4}R$



18. B

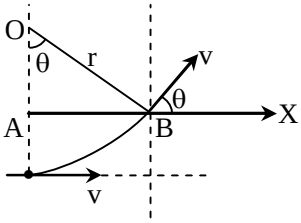
Sol. $F \propto \frac{x}{(x^2+a^2)^{3/2}}$; $a = \frac{F}{m}$



19. D

Sol. In ΔOAB

AB = L = constant



$OB = r = \frac{mv}{qB}$

$\sin\theta = \frac{L}{r} \Rightarrow L = r\sin\theta$ = constant

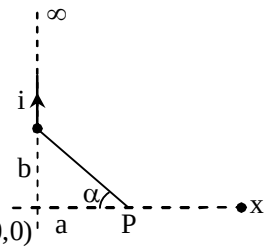
or $v \sin\theta$ = constant

$v_0 \sin 60^\circ = v' \sin 30^\circ$

$v' = \sqrt{3}v_0$

20. B

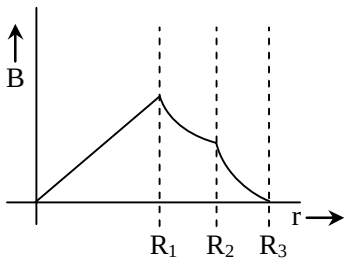
Sol. $B = \frac{\mu_0}{4\pi}\frac{i}{a}(1 - \sin\alpha)$



$B = \frac{\mu_0}{4\pi}\frac{i}{a}\left(1 - \frac{b}{\sqrt{a^2+b^2}}\right)$

21. C

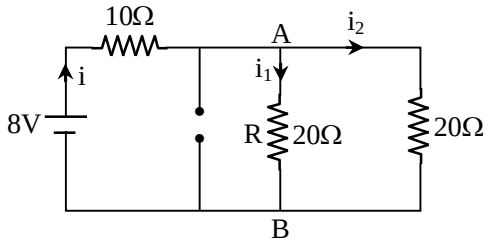
Sol. From ampere's law the field at the axis is zero. From x = zero to R_1, the field increases linearly as the charge enclosed increases. From x = R_1 to R_2 and from x = R_2 to R_3, the field decreases hyperbolically but with different slopes as the media are different. Hence the required graph is.



$$B=\begin{cases}\dfrac{\mu_0}{2\pi}\dfrac{ir}{R_1^2} & \text{for} \quad r \le R_1 \\ \dfrac{\mu_0}{2\pi}\dfrac{i}{r} & \text{for} \quad R_1 \le 2r \le R_2 \\ 0 & \text{for} \quad r > R_3\end{cases}$$

22. A

Sol. The given circuit can be simplified as



at steady state

$R_{eq} = 20\Omega$

$i = \dfrac{8}{20} = 0.4$ A

$i_1 = 0.2$A

$V_{AB} = 4$V

charge on capacitor $q = CV = 0.1 \times 4$

$= 0.4\ \mu C$

23. A

Sol. We have

$$N = N_0\left(\frac{1}{2}\right)^n$$

Where n is no. of half lives i.e., $\dfrac{1}{2}$

Hence $N = \dfrac{N_0}{\sqrt{2}}$ or $\dfrac{N}{N_0} = \dfrac{1}{\sqrt{2}}$

24. B

Sol. Reducible in exact form

25. D

Sol. $B_n = \dfrac{\mu_0 I_n}{2r_n}$

or $B_n \propto \dfrac{I_n}{r_n} \propto \dfrac{f_n}{r_n} \propto \dfrac{v_n}{r_n^2}$

$\Rightarrow B_n \propto \dfrac{z/n}{(n^2/z)^2} \propto \dfrac{z^3}{n^5}$

26. C

Sol. For M_1 : $u = -10$ cm

$f = -15$ cm

$n_0 = 2$

Using mirror formula v = 30 cm
and n_I = 6 cm
The image formed by plane mirror is at 70 cm below the principle axis and 70 + 6 – 30 = 46 cm from concave mirror.
So, coordinates of second image w.r.t.
P = (– 46, – 70)

27. C

Sol. For upper segment,

$$\frac{1}{f} = \frac{\mu - 1}{R}$$

For lower segment,

$$\frac{1}{f} = \frac{2(\mu - 1)}{R}$$

28. A

Sol. $N = \frac{X}{\lambda} = \frac{X}{\frac{\ell n(2)}{Y}} = \frac{XY}{\ell n(2)}$

29. B

Sol. For reflection through M_1
u = –15 cm
f = –10 cm

$$\frac{1}{v} = -\frac{1}{10} + \frac{1}{15} \Rightarrow v = -30 \text{ cm}$$

For reflection through M_2
u = 10 cm
f = –10 cm
$\Rightarrow$ v = – cm (from mirror formula)

Now magnification, $m = -\frac{v}{u} = \frac{5}{10} = \frac{1}{2}$

So, finally formed image will be at the distance of $\frac{3}{2}$ cm from the line AB.

30. B

Sol. For any outside stationary observer (Paraxial rays approx) the fish will be appeared at 15 cm from surface and moving with speed of 4 cm/sec in upward direction

Now, use $\vec{v}_{1m} = -m^2 v_{OM}$

where $m = \frac{-50}{-50+30} = \frac{5}{2}$

$\Rightarrow v_I - (-4) = -\frac{25}{4}\ [4 - (-4)]$

$\Rightarrow v_I + 4 = -50$

$\Rightarrow v_I = -54$ m/sec

(–ve sign showing downward direction)

Chemistry

PART – II

SECTION – A

31. D

Sol. $NH_4SH(s) \rightleftharpoons NH_3(g) + H_2S(g)$

$t = 0$	0	0.5 atm	
$t = teq$	0	$0.5 + P$	P

Total Pressure=0.5+P+P=0.84

P=0.17 atm

$K_P = P_{NH_3} \cdot P_{H_2S} = (0.5 + 0.17) \times (0.17) = 0.1139\, atm$

32. C

Sol. $Mg_2C_3 + H_2O \rightarrow Mg(OH)_2 + C_3H_4$

Propyne

33. B

Sol. $2BeCl_2 + LiAlH_4 \rightarrow 2BeH_2 + LiCl + AlCl_3$

34. B

Sol. $m \cdot moles\ of\ HCl = 1 \times 0.1 = 0.1$

$now\ Vol\ of\ solution = 1 + 99 = 100\, cc$

$Molarity\ of\ HCl = \frac{0.1}{100} = 10^{-3}$

$pH = -\log\left[H^+\right] = -\log\left(10^{-3}\right) = 3$

35. B

Sol. $M_1V_1 + M_2V_2 = M(V_1 + V_2)$

$\frac{1}{200} \times 10 + \frac{1}{200} \times 40 = M(10 + 40)$

$M = \frac{1}{200}$

$H_2SO_4 \rightarrow 2H^+ + SO_4^{-2}$

$\left[H^+\right] = 2 \times \frac{1}{200} = 10^{-2}$

$pH = -\log\left[H^+\right] = -\log\left(10^{-2}\right) = 2$

36. A

Sol. Mg can form Complex (Mg-EDTA Complex)

37. D

Sol. Due to their covalent character

38. A

Sol. $P_4 + 3NaOH + 3H_2O \rightarrow 3NaH_2PO_2 + PH_3$

39. A

Sol. $H_3BO_3 \xrightarrow{T_1>170^\circ} HBO_2 + H_2O$

meta boric acid

$4HBO_2 \xrightarrow{T_2>300^\circ C} H_2O + H_2B_4O_7$

Tetra boric acid

$H_2B_4O_7 \xrightarrow{red\ hot} 2B_2O_3 + H_2O$

40. A

Sol. Information base

41. C

Sol. $R = K[A]^o$

$10^{-2} = K \times (0.1)^o$

$K = 10^{-2}$

$A_t = A_o - Kt$

$= 0.1 - 10^{-2} \times \frac{60}{60}$

$A_t = 0.1 - 0.01$

$[B] = 2 \times 0.01 = 0.02$

42. A

Sol.

	$N_2O_4 \rightleftharpoons$	$2NO_2$
$t = 0$	1	0
$t = teq$	$1-\alpha$	2α

$M_{eq} = \frac{1 \times M_{N_2O_4}}{1-\alpha+2\alpha} = \frac{92}{1+\alpha}$

$2 \times VD = \frac{92}{1+\alpha}$

$\alpha = 0.533$

43. D

Sol.

	$2HI \rightleftharpoons$	H_2 +	I_2
$t = 0$	1	0	0
$t = teq$	$1-\alpha$	$\frac{\alpha}{2}$	$\frac{\alpha}{2}$

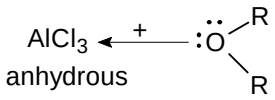
$K_P = \frac{P_{H_2} \cdot P_{I_2}}{(P_{HI})^2} = \frac{\left(\frac{\alpha}{2(1+\alpha)}P\right)\left(\frac{\alpha}{2(1+\alpha)}P\right)}{\left(\frac{1-\alpha}{1+\alpha}P\right)^2}$

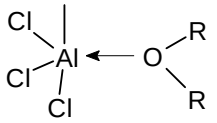
$K_P = \frac{\alpha^2}{4(1-\alpha)^2}$

$\alpha = \frac{2\sqrt{K_P}}{1+2\sqrt{K_P}}$

44. D
Sol. It is due to solvated electron.

45. B
Sol.





46. B
Sol. $R = K[A]^m[B]^n$

$10^{-3} = K\left(1.6\times10^{-3}\right)^m\left(5\times10^{-2}\right)^n$

$3.2\times10^{-3} = K\left(3.2\times10^{-3}\right)^m\left(5\times10^{-2}\right)^n$

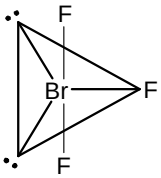
$1.6\times10^{-3} = K\left(10^{-1}\right)^m\left(2\times10^{-3}\right)^n$

$3.2\times10^{-3} = K\left(10^{-1}\right)^m\left(8\times10^{-3}\right)^n$

Solving $R = K[A]^2[B]$

47. C
Sol. $KCN \longrightarrow K^+ + {}^-C \equiv N$

48. A
Sol.



SP^3d hybridisation
T-shaped

49. B
Sol. $Cl_2O > H_2O > F_2O$ (bond angle)

50. C
Sol. Electron affinity S>Se>O

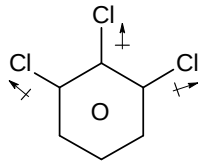
51. D
Sol. $Cl < Mg < Ca$

52. D
Sol. $r_n = \dfrac{n^2}{z} r_o$

53. B
Sol. $\dfrac{1}{\lambda} = RZ^2\left(\dfrac{1}{n_1^{\,2}} - \dfrac{1}{n_2^{\,2}}\right)$

54. A

Sol.



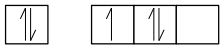
55. D

Sol. Bond order $H_2O_2 < O_3 < O_2$

Bond length $H_2O_2 > O_3 > O_2$

56. D

Sol.



57. B

Sol. Non metal oxide are acidic metal oxide are basic

58. C

Sol. $Hyd \propto \frac{q_1 q_2}{r}$

59. C

Sol. $K_P = K_c \cdot (RT)^{\Delta ng}$

60. C

Sol. $t_{1/2} \propto (a)^{1-n}$

hence $n = 0$

$t_{1/2} \propto a; '0'$ *order reaction*

$Rate = -\frac{d[n]}{dt} = K[A]^o = const$

Mathematics

PART – III

SECTION – A

61. B

Sol. Take log both sides, then differentiate

62. C

63. D

Sol. $f(n)=\sum(n+1)(n+\omega)(n+\omega^2)=\sum(n+1)(n^2-n+1)=\sum(n^3+1)$

64. C

Sol. $|z^2|+2(z+\bar{z})-5=0: centre=(-2,0)\, radius=\sqrt{4+5}=3$

$\therefore \quad z=-2+3e^{i\theta} \Rightarrow z+3+2i=L+2i+3e^{i\theta}$

65. D

Sol. $f(f(x))=\lambda(x^7+2x), \lambda\neq 0 \quad f'(f(x))\times f'(x)=\lambda(7x^6+2)$

66. A

Sol. For any one diagonal, there are 7 lines skew diagonal

$\therefore No=\frac{12\times 5}{2}=30$

67. C

Sol. Using truthtable

68. B

Sol. $\sum x_i=48\times 200; \sum(x-\mu)^2=200\times 9$

$\Rightarrow \sum x^2-n\mu^2=200\times 9 \Rightarrow \sum x^2=462600$

69. D

Sol. $\cot^{-1}x>0 \ \& \ \cos^{-1}x>0$

$\therefore$ No solution

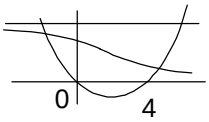
70. D

Sol. $A=A^T \Rightarrow x^2+y^2+z^2=1; xy+yz+zx=0 \Rightarrow x+y+z=1$

$\therefore x^3+y^3+z^3=4$

71. C

Sol.



72. D

Sol. $\frac{x^2}{18}-\frac{y^2}{9}=1$

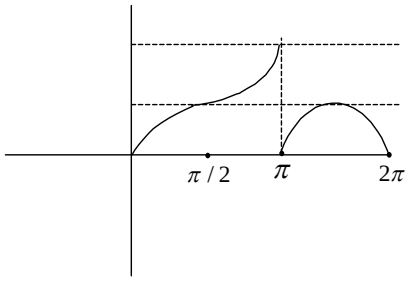
Equation of Tangent : y = -x + k

$\therefore\ K^2 = 18(-1)^2 - 9$

$K = \pm 3$

73. B

Sol. $f(x) = \begin{cases} \sin x & 0 \le x \le \pi/2 \\ 2 - \sin x & \frac{\pi}{2} < x \le \pi \\ -\sin x & \pi < x < 2\pi \end{cases}$



Points of discontinuity x = π (n = 1)

Points of non-dertvability x = π (m = 1)n + m = 2

74. A

Sol. $h(x) = f^2(x) + \left(f'(x)\right)^2$

$h'(x) = 2f(x)f'(x) + 2f'(x).f''(x)$

$= 2f'(x)\left(f(x) + f''(x)\right)$

$= -2xg(x)\left(f'(x)\right)^2$

h'(x) changes from positive to negative about x = 0 so x = 0 is a point of maxima

75. C

Sol. $\alpha + \sin\beta = 2018$

$\alpha + 2018\cos\beta = 2017$

satisfies

$if\ \alpha = 2017\ and\ \beta = \frac{\pi}{2}$

$[\alpha + \beta] = \left[2017 + \frac{\pi}{2}\right] = 2018$

76. B

Sol. $\frac{x^2}{8/9} - \frac{y^2}{8/9} = 1$

$Equation\ of\ Tangent: y = mx \pm \sqrt{\frac{8}{9}\left(m^2 - 1\right)} ----(i)$

$y^2 = 32x$

$Equation\ of\ Tangent: y = mx + \frac{8}{m} ------(ii)$

on comparing equation (i) and (ii)

$m = \pm 3$

putting m in equation(ii)

$y = 3x + 8/3$ *or* $y = -3x - 8/3$

$a = \frac{-8}{9}$

77. D

Sol. $\left(x^2 + y^2 + 8x + 8y + 16\right) + \lambda\left(x + y + 12\right) = 0$

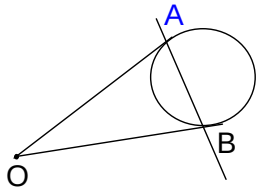
Equation of line AB = x + y + 12 = 0 ----------(i)

Equation of chord of contact:

$\left(8 + \lambda\right)x + \left(8 + \lambda\right)y + 2\left(16 + 12\lambda\right) = 0 - - - -$(ii)

On comparing equation (i) and (ii)

$\lambda = \frac{16}{3}$



x + y + 12 = 0

78. D

Sol.

2
1
1
2

$= \left(3c_2 . 1.1.3c_2\right)6!$

Total = 10800

1
1
1
3

$= \left(3c_1 . 1.1.1\right)6!$

3
1
1
1

$= \left(1.1.1.3c_1\right)6!$

79. B

80. D

Sol. On comparing coefficients

$a_0 = 1$ $- - - - -$(i) (constant term)

$n - 2 = a_1 - - - - -(ii)$(coeff .of x)

$5 + {}^n c_2 - 2n = a_2 - - - -(iii)\left(coeff .of\ x^2\right)$

On solving (ii) and (iii)

n = 6

$\therefore a_0 + n = 7$

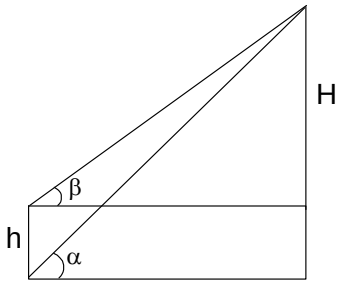
81. D

Sol. $V_1 = \left[\vec{\alpha} \quad \vec{\beta} \quad \vec{\gamma}\right]$

$V_2 = \left[\vec{a} \quad \vec{b} \quad \vec{c}\right]$

82. B

Sol.



83. A

Sol. $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}=\alpha$

Point P on line

$P(2\alpha+1,3\alpha+2,4\alpha+3)$

$\frac{x-4}{1}=\frac{y-6}{2}=\frac{z+\lambda}{2}=\beta$

Any point Q

$Q(4+\beta,6+2\beta,2\beta-\lambda)$

For certain value of α & β P & Q represents same points

$2\alpha+1=4+\beta$ ----- (i)

$3\alpha+2=6+2\beta$ ---- (ii)

$4\alpha+3=2\beta-\lambda$ ---- (iii)

From (i) & (ii)

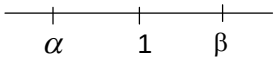
α=2 & β = 1 now from (iii)

$\lambda=-9$

84. A

85. B

Sol.



$(k^2-4k+5)(k^2-4k+5+(2k-1)-3k)<0$

86. C

Sol. 1^{∞} form

87. A

Sol. $x \Rightarrow a+b-x$

88. B

89. D

90. C